CHAPTER 23 · DERIVATIONS

GEOMETRICAL OPTICS

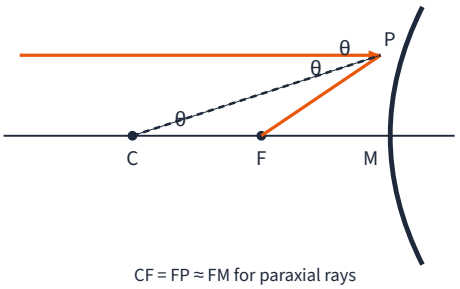
2 derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 FOCAL LENGTH OF A SPHERICAL MIRROR

SYLLABUS 23.1(A)

RESULT $f = \frac{r}{2}$

1. A ray parallel to the axis strikes a concave mirror at P. The normal at P passes through the centre of curvature C.
2. Angle of incidence = angle of reflection = θ. The parallel ray makes angle θ with CP too (alternate angles), so triangle CFP has two equal angles.
3. CF = FP (isosceles triangle).
4. For paraxial rays P is close to the pole M, so FP ≈ FM. Then CM = CF + FM ≈ 2FM.
5. r = 2f, so $\mathbf{f = \frac{r}{2}}$.



CF = FP ≈ FM for paraxial rays

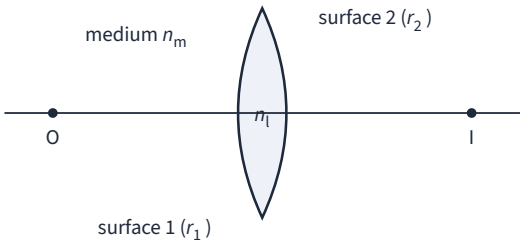
Use the sign convention consistently.

2 THIN LENS FORMULA AND LENSMAKER'S EQUATION

SYLLABUS 23.3(A)

RESULT $\frac{1}{u} + \frac{1}{v} = \frac{1}{f}; \frac{1}{f_m} = (\frac{n_l}{n_m} - 1)(\frac{1}{r_1} - \frac{1}{r_2})$

1. Lens of refractive index n_l in a medium n_m. Use $\frac{n_1}{u} + \frac{n_2}{v} = \frac{n_2 - n_1}{r}$ at each surface.
2. Surface 1 (medium → lens): $\frac{n_m}{u} + \frac{n_l}{v_1} = \frac{n_l - n_m}{r_1}$ … (1)
3. The image I_1 is the object for surface 2. The lens is thin, so it is a virtual object at $u_2 = -v_1$.
4. Surface 2 (lens → medium): $-\frac{n_l}{v_1} + \frac{n_m}{v} = \frac{n_m - n_l}{r_2}$ … (2)
5. Add (1) and (2): $\frac{n_m}{u} + \frac{n_m}{v} = (n_l - n_m)(\frac{1}{r_1} - \frac{1}{r_2})$.
6. Divide by n_m: $\frac{1}{u} + \frac{1}{v} = (\frac{n_l}{n_m} - 1)(\frac{1}{r_1} - \frac{1}{r_2})$.
7. Object at infinity: $v = f_m$, so $\mathbf{\frac{1}{f_m} = (\frac{n_l}{n_m} - 1)(\frac{1}{r_1} - \frac{1}{r_2})}$ (lensmaker's equation), and therefore $\mathbf{\frac{1}{u} + \frac{1}{v} = \frac{1}{f}}$.



In this form r is positive when the centre of curvature is on the side the light goes to, so for a biconvex lens r_2 is negative. Many books write $(\frac{1}{r_1} + \frac{1}{r_2})$ with both radii of a convex lens positive; both give the same f.